

- 10** Jade is a parkour enthusiast who enjoys jumping between built structures or obstacles. One day, Jade decides to jump from one 2-metre-high structure to another of the same height. The path of Jade's jump can be modeled by the quadratic equation:

$$h = -\frac{5}{2}t^2 + 5t + 1 \text{ where } h \text{ is the height in metres above the ground after } t \text{ seconds.}$$

- (a)** Rewrite h in the form $b - k(t - a)^2$. [3]

- (b)** Hence determine
(i) the maximum height that Jade reaches during the jump, [1]

- (ii)** the time at which this maximum height occurs. [1]

- (c)** Find the total time Jade spends in the air before landing on the other building. [3]